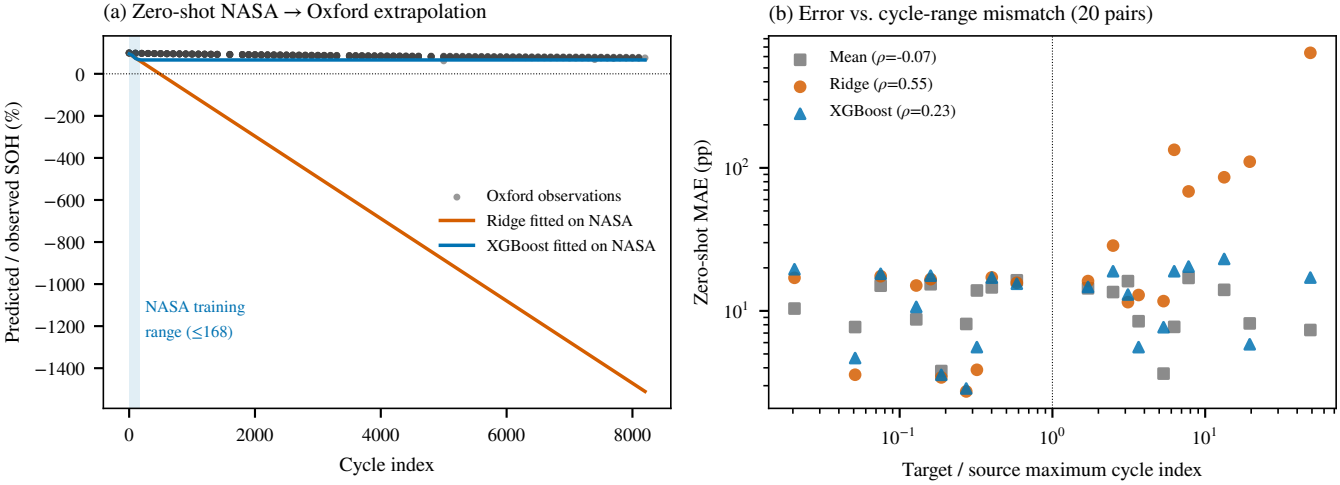
(a) Zero-shot NASA → Oxford extrapolation
Predicted / observed SOH (%)
0
−200
−400
−600
−800
−1000
−1200
−1400
0
2000
4000
6000
8000
Cycle index
Oxford observations
Ridge fitted on NASA
XGBoost fitted on NASA
NASA training
range (≤168)
(b) Error vs. cycle-range mismatch (20 pairs)
Mean (ρ=-0.07)
Ridge (ρ=0.55)
XGBoost (ρ=0.23)
Zero-shot MAE (pp)
10²
10¹
10⁻¹
10⁰
10¹
Target / source maximum cycle index